Abdelhakim Amer

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MSc. in Mechanical Engineering

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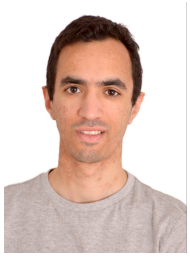
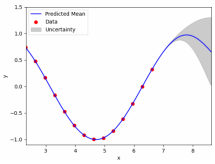
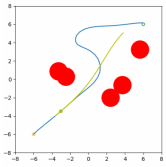
Profile Summary



I am passionate about robotics, autonomous systems, learning-based control, and field robotics for industrial applications. My research focuses on leveraging **machine learning** and **optimization** to improve robot trajectory tracking and enhance control, planning, and dynamics modeling in autonomous mobile robots. I'm driven by the challenge of advancing autonomous systems and bridging **theory** with practical solutions for **real-world applications**. My work includes:



Gaussian Processes



Model Predictive Control

Advanced Simulation

Academics

I hold a bachelor's in Mechanical Engineering (with High Honors), and a master's in High Tech Mechatronics Systems from Delft University of Technology, specializing in numerical methods, modeling, and control. I have been a visiting scholar at the German Center for Artificial Intelligence (DFKI). I am currently pursuing an Industrial PhD in learning-based optimal control at Aarhus University.

2022 - now	EIVA A/S, Denmark	**PhD**, Robotics and A.I
2024 (1 month)	DFKI, Germany	**Visiting Scholar**, Underwater Robotics Research Group
2023 (1 month)	University of Paderborn, Germany	**Visiting Scholar**, Automatic Control Research Group
2018 - 2021	Delft University of Technology, The Netherlands	**Master**, Mechanical Engineering
2013 - 2018	The American University in Cairo, Egypt	**Bachelor**, Mechanical Engineering (High Honors)

Work Experience

Prior to my PhD, I completed a 3-month internship at Royal IHC, where I worked on developing algorithms for path planning and control. After graduation, I spent a year at the AIR Lab at Aarhus University, working on an autonomous wind turbine inspection project in collaboration with Upteko, where I gained experience in software and hardware development for drones. I then began my industrial PhD at EIVA, where I am finalizing my research in learning-based optimal control, having developed state-of-the-art methods for planning and control of autonomous underwater vehicles, with a focus on addressing model uncertainties.

EIVA	2022 - 2025	EIVA A/S	PhD Researcher in Robotics and AI
	2021 - 2025	Aarhus University	Graduate Teaching Assistant in Control and Robotics
UPTEKO	2021 - 2022	Upteko Aps	R&D Engineer
AiRLab	2021 - 2022	AIR Lab, Aarhus University	Research Assistant in Autonomous Systems
IHC	2020 (3 months)	Royal IHC	Dynamics and Control Graduate Intern

Programming Experience

In recent years, I have primarily developed in Python and C++ for robotics applications, particularly in learning-based control and simulation. I have extensive experience with the ROS framework for integrating robotic systems. Additionally, I have worked with Unreal Engine for developing high-fidelity simulations.

Python	C++	ROS	Unreal Engine	MATLAB
★★★★★	★★★★☆	★★★★★	★★★☆☆	★★★★☆

Teaching and Supervision

As a Teaching Assistant at Aarhus University, I contributed to the following courses, providing support to students and assisting with course material:

- Autonomous Mobile Robots
- Control Theory

Additionally, I have supervised several Bachelor's and Master's theses, guiding students through projects in areas such as ROTV design, underwater vehicle modeling, and drone design.

Fall 2024 MSc Thesis: Underwater Vehicle Modeling and Control using PINNs

Fall 2023 R&D Project: 3-D printing ship propeller cleaning ROV design

Spring 2023 BSc Thesis: Modeling and Control of a Seabed Visualizing ROTV using LQR

Fall 2022 BSc Project: Mini-Inspection Drone Design, Build and Control

Awards

I have received the following awards during my studies:

Innovation Fund Denmark Funding – 3-year project on A.I.-driven optimal control for inspection and cleaning project (2022), amount: 1.3M DKK.

Academic Achievement Scholarship (2013-2018), The American University in Cairo – Awarded for outstanding academic performance during undergraduate studies.

Publications

The following list of publications represents the outcomes of my research from my Master's thesis to the present. Throughout my research projects, I have collaborated with researchers from various research institutions and companies across the world, including Denmark, Germany, Finland, China, and the USA.

1. **Amer, A.**, Falsegar, D., Brodskiy, Y., & Sarabakha, A. (2025). *Modelling of Underwater Vehicles using Physics-Informed Neural Networks with Control*. Submitted to the International Joint Conference on Neural Networks (IJCNN), under review.

2. **Amer, A.**, Mehndiratta, M., Brodskiy, Y., & Kayacan, E. (2025). *Empowering Autonomous Underwater Vehicles Using Learning-based Model Predictive Control With Dynamic Forgetting Gaussian Processes*. IEEE Transactions on Control Systems Technology.

3. Liang, W., **Amer, A.**, Mehndiratta, M., Chen, Z., Yao, B., & Kayacan, E. (2025). *Adaptive Robust Control Integrated With Gaussian Processes for Quadrotors: Enhanced Accuracy, Fault Tolerance and Anti-Disturbance*. IEEE Transactions on Systems, Man, and Cybernetics: Systems.

4. **Amer, A.**, Mehndiratta, M., Sejersen, J.L.F., Pham, H.X., & Kayacan, E. (2023). *Visual Tracking Nonlinear Model Predictive Control Method for Autonomous Wind Turbine Inspection*. 2023 21st International Conference on Advanced Robotics (ICAR), 431-438.

5. **Amer, A.**, Álvarez-Tuñón, O., Uğurlu, H.İ., Sejersen, J.L.F., Brodskiy, Y., & Kayacan, E. (2023). *UNav-Sim: A visually realistic underwater robotics simulator and synthetic data-generation framework*. 2023 21st International Conference on Advanced Robotics (ICAR), 570-576.

6. **Amer, A.**, Álvarez-Tuñón, O., Falsegar, D., Brodskiy, Y., & Kayacan, E. (2023). *MUDROV: A modular underwater defouling ROV for ship propeller cleaning*. Advanced Marine Robotics Workshop, International Conference on Intelligent Robots and Systems (IROS) 2023.

7. **Amer, A.** (2021). *Shape Optimization of a Flapping Wing for the Atalanta Project FWMAV* (Master's thesis). Delft University of Technology.

Best regards,
Abdelhakim Amer